

AURA AI ASSISTANT — FULL SYSTEM REPORT (STEP 0 → PRODUCTION)

1. INTRODUCTION

The AURA system (Adaptive Unified Real Estate Intelligence Engine) is designed to evolve from a **data-driven analytics platform (AURA V5)** into a **fully autonomous AI decision infrastructure (AURA V6)**.

This transformation is not simply the addition of a chatbot, but the creation of a **reasoning engine capable of interacting with structured real estate data, generating insights, and supporting high-stakes decision-making**.

Unlike traditional AI systems, AURA does not rely on static predictions. Instead, it integrates:

- Large-scale structured datasets (1.25M+ rows)
- Real-time analytical tools
- A locally deployed large language model (LLM)

The result is a **sovereign AI assistant** that understands, queries, analyzes, and explains real estate intelligence.

2. STEP 0 — STRATEGIC FOUNDATION (COMPLETED)

The foundation of AURA V5 has already been successfully established, including:

- Construction of a **fused dataset (1.25M rows, 174 features)**
- Development of a **data fusion pipeline**
- Integration of multiple data categories (market, socio-economic, spatial)
- Initial exploration of LLM-based intelligence systems

This phase ensures that the system has **high-quality, structured, and decision-relevant data**, which is the core asset of AURA.



3. SYSTEM EVOLUTION

The system evolves as follows:

AURA V5 → Data Intelligence Platform

AURA V6 → AI Decision Engine

In AURA V5, intelligence is derived from models and dashboards.

In AURA V6, intelligence is generated dynamically through **LLM reasoning over data**.



4. GLOBAL ARCHITECTURE

The architecture of the AURA AI Assistant is structured as:

User / Admin Interface



FastAPI Backend



AURA AI Orchestrator



Local LLM (Qwen)



Tool Layer (SQL + Analytics)



AURA V5 Dataset

Key Principle

The LLM does **not store knowledge internally**, but acts as an **orchestrator** that:

- Understands user intent
- Translates it into structured operations (SQL / analysis)
- Retrieves real data
- Generates explanations

This guarantees **accuracy, scalability, and explainability**

5. PHASE 1 — LLM DEPLOYMENT (LOCAL BRAIN)

Objective

Deploy a local LLM capable of reasoning and structured output generation.

Implementation

- Install runtime (Ollama or Transformers)
- Load a quantized model (e.g., Qwen 7B)
- Validate capabilities:
 - Logical reasoning
 - SQL generation
 - Structured JSON outputs

Outcome

A fully functional **local AI brain**, independent from external APIs.

6. PHASE 2 — TOOL LAYER (CORE INTELLIGENCE)

Objective

Connect the LLM to real-world data through executable tools.

Components

1. SQL Engine

```
def query_aura(sql):  
    return db.execute(sql)
```

2. Analytics Engine

```
def analyze_market(data):  
    return trend, volatility, signals
```

3. Data Access Layer

```
def get_grid_data(grid_id, year, quarter):  
    return df_filtered
```

Core Mechanism

LLM → generates SQL → tool executes → returns data → LLM explains

This approach ensures **data-grounded reasoning instead of hallucination**

7. PHASE 3 — LLM ORCHESTRATION (AURA CORE ENGINE)

Objective

Transform the LLM into an intelligent decision engine.

Implementation

- Define system prompts (AURA rules & logic)
- Enable structured outputs (JSON-based)
- Implement tool-calling routing

Execution Flow

User Query
→ LLM interprets intent
→ selects tool
→ executes query
→ receives data
→ generates explanation

Example

User:

“Best investment area in Lisbon?”

System:

- Queries multiple datasets (pricing, demand, risk)
- Computes ROI indicators
- Returns a justified recommendation



8. PHASE 4 — DOMAIN ADAPTATION



Objective

Align the LLM with real estate intelligence logic.



Methods

Prompt Engineering (Primary)

- Encode domain rules
- Define decision logic
- Guide reasoning patterns

Optional Fine-Tuning (LoRA / PEFT)

- Improve domain-specific performance
- Not mandatory for MVP



Outcome

LLM behaves as a **real estate expert**, not a generic assistant.



9. PHASE 5 — EXPLAINABILITY SYSTEM



Objective

Ensure transparency and trust in AI decisions.



Components

- SHAP → feature importance (admin view)
- LLM explanations → human-readable insights



Example Output

“The price increase is driven by high demand, limited supply, and strong demographic growth.”

This aligns with modern AI regulations requiring explainability

10. PHASE 6 — OPTIONAL DUAL MODEL ARCHITECTURE

Architecture

Lightweight Model (Gemma) → user interaction

Advanced Model (Qwen) → reasoning engine

Purpose

- Reduce cost
- Improve scalability
- Support mobile environments

This phase is optional for internship scope.

11. PHASE 7 — DEPLOYMENT (MVP SYSTEM)

Objective

Deliver a usable AI assistant.

Stack

- FastAPI (backend)
- Local LLM (Qwen)
- Database (MySQL / CSV)
- API endpoints or minimal UI

Flow

Frontend → API → LLM → Tools → Dataset → Response

12. PHASE 8 — OPTIMIZATION

Improvements

- Reduce latency
 - Optimize prompts
 - Add caching
 - Improve throughput
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13. PHASE 9 — FUTURE EVOLUTION (AURA V6+)

Upgrade Path

- Larger models (MoE architectures)
 - Real-time data pipelines
 - Multi-agent AI systems
 - Cloud-native deployment
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14. IMPLEMENTATION TIMELINE

Week	Objective
1	LLM local deployment
2	Tool layer (SQL engine)
3	LLM + dataset integration
4	Reasoning + explanations
5	API + demo
6	Optimization + presentation

15. FINAL SYSTEM CAPABILITIES

The final AURA AI Assistant will:

1. Understand complex user queries
 2. Translate them into structured data operations
 3. Retrieve real data from the dataset
 4. Perform analytical reasoning
 5. Provide clear, explainable insights
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FINAL CONCLUSION

The AURA transformation represents a shift from:

Static Data Platform → Dynamic AI Decision System

By integrating:

- A structured data infrastructure
- A tool-based execution layer
- A locally deployed LLM

AURA becomes a **sovereign, intelligent system capable of reasoning over real-world data and delivering actionable insights.**

This architecture ensures:

- Independence from external APIs
 - High accuracy through data grounding
 - Full control over AI behavior
 - Scalability toward enterprise deployment
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FINAL STATEMENT

AURA is not a chatbot.

It is:

An AI-powered real estate decision infrastructure

If you want next step:

👉 I can convert this into:

- **LaTeX PFE report (perfect format)**
- **Architecture diagrams**
- **Code-level backend (FastAPI + LLM + SQL)**

[AURA AI — Phase 9 Completion Report](#)